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```
clc;
clear;
close all;
```

===== PARAMETERS

=====

```
Nfft = 64;
Nofdm = 100;
CP = 16;
```

```
SNR_dB = 0:5:25;
```

```
BER_QPSK = zeros(size(SNR_dB));
BER_16QAM = zeros(size(SNR_dB));
```

=====

=====

QPSK

=====

```
M = 4;
k = log2(M);
```

```
Nbits = Nofdm * Nfft * k;
```

```
txBits = randi([0 1], Nbits, 1);
```

```
txSymbols = qammod(txBits, M, ...
    'InputType', 'bit', ...
    'UnitAveragePower', true);
```

```
txMatrix = reshape(txSymbols, Nfft, Nofdm);
```

```
txOFDM = ifft(txMatrix, Nfft);
```

```
% Add cyclic prefix
CP_data = txOFDM(end-CP+1:end, :);
txOFDM_CP = [CP_data; txOFDM];
```

QPSK SNR LOOP

```
for i = 1:length(SNR_dB)

    % AWGN channel
    rx = awgn(txOFDM_CP, SNR_dB(i), 'measured');

    % Remove CP
    rx = rx(CP+1:end, :);

    % FFT
    rxSymbols = fft(rx, Nfft);

    % Serial conversion
    rxSymbols = rxSymbols(:);

    % Demodulation
    rxBits = qamdemod(rxSymbols, M, ...
        'OutputType', 'bit', ...
        'UnitAveragePower', true);

    % BER
    [~, BER_QPSK(i)] = biterr(txBits, rxBits);

end
```

```
=====
=====
```

16-QAM

```
=====

M = 16;
k = log2(M);

Nbits = Nofdm * Nfft * k;

txBits = randi([0 1], Nbits, 1);

txSymbols = qammod(txBits, M, ...
    'InputType', 'bit', ...
    'UnitAveragePower', true);

txMatrix = reshape(txSymbols, Nfft, Nofdm);

txOFDM = ifft(txMatrix, Nfft);
```

```
% Add cyclic prefix
CP_data = txOFDM(end-CP+1:end, :);
txOFDM_CP = [CP_data; txOFDM];
```

16-QAM SNR LOOP

```
for i = 1:length(SNR_dB)

    % AWGN channel
    rx = awgn(txOFDM_CP, SNR_dB(i), 'measured');

    % Remove CP
    rx = rx(CP+1:end, :);

    % FFT
    rxSymbols = fft(rx, Nfft);

    % Serial conversion
    rxSymbols = rxSymbols(:);

    % Demodulation
    rxBits = qamdemod(rxSymbols, M, ...
        'OutputType', 'bit', ...
        'UnitAveragePower', true);

    % BER
    [~, BER_16QAM(i)] = biterr(txBits, rxBits);

end
```

===== COMPARISON

=====

```
figure;

semilogy(SNR_dB, BER_QPSK, '-o', ...
    'LineWidth', 1.5);

hold on;

semilogy(SNR_dB, BER_16QAM, '-s', ...
    'LineWidth', 1.5);

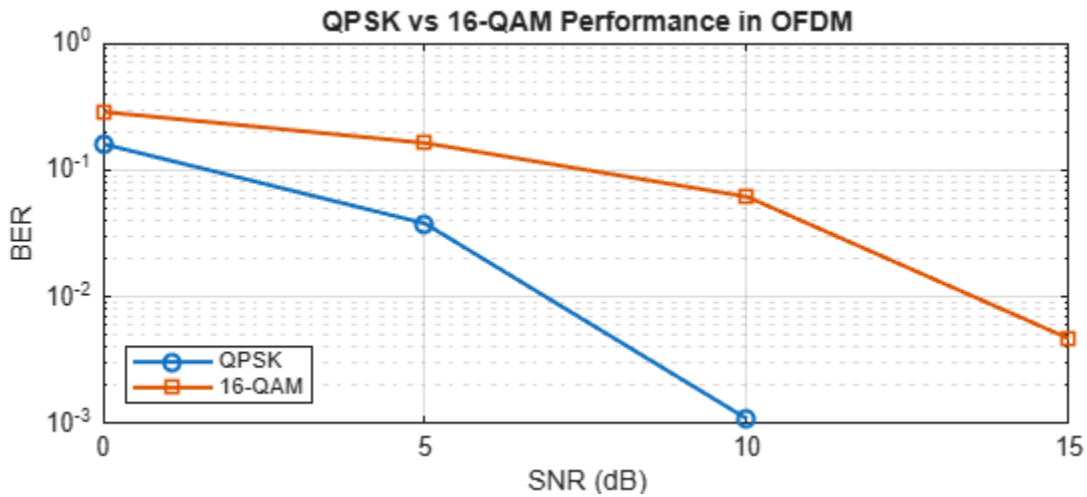
grid on;

xlabel('SNR (dB)');
ylabel('BER');

title('QPSK vs 16-QAM Performance in OFDM');

legend('QPSK', ...
```

```
'16-QAM', ...
'Location', 'southwest');
```



===== DISPLAY RESULTS =====

```
fprintf('\n');
fprintf('SNR(dB)          QPSK BER          16-QAM BER\n');
fprintf('-----\n');

for i = 1:length(SNR_dB)

    fprintf('%2d          %.6e          %.6e\n', ...
            SNR_dB(i), ...
            BER_QPSK(i), ...
            BER_16QAM(i));

end
```

SNR(dB)	QPSK BER	16-QAM BER
0	1.604687e-01	2.872266e-01
5	3.789062e-02	1.631641e-01
10	1.093750e-03	6.156250e-02
15	0.000000e+00	4.687500e-03
20	0.000000e+00	0.000000e+00
25	0.000000e+00	0.000000e+00

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